

WINDOWS SERVER ADMINISTRATION

Active Directory, Group Policy, DNS, DHCP and File Server

Environment:

VirtualBox

Server:

Windows Server

Client:

Windows 10 Pro

Domain:

Company.local

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06-10-26

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PROJECT OVERVIEW

This Project involved building and administering a small windows server environment in VirtualBox to simulate a real-world IT infrastructure.

The lab was designed to develop practical skills in Windows Server administration, Active Directory, Group Policy, DNS, DHCP, domain management, file sharing, access control, and user account administration.

The environment consisted of a Windows Server domain controller and a Windows 10 Pro client joined to the company.local domain. Active Directory was used to organize users and departments, while Group Policy and security permissions were configured to manage and control access to company resources.

The completed environment was tested to verify domain connectivity, policy application, DNS resolution, IP address assignment, file-share accessibility, and department-based access permissions.

Key Objectives

- Deploy and configure a Windows Server domain environment.
- Configure Active Directory users, organizational units, and security groups.
- Join a Windows 10 Pro workstation to the domain.
- Configure and verify Group Policy.
- Configure DNS and DHCP services.
- Create and secure an SMB file share.
- Implement and test Share and NTFS permissions.
- Perform basic Active Directory user account administration.
- Verify that the configured environment operates as intended.

1. LAB ENVIRONMENT

The lab was built using Oracle VirtualBox with one Windows Server virtual machine and one Windows 10 Pro client virtual machine.

VIRTUAL MACHINES

Component	Configuration
Server	Windows Server
Server Name	Sv02
Client	Windows 10 Pro
Client Name	CLL01
Virtualization Platform	Oracle VirtualBox
Active Directory Domain	company.local

NETWORK ENVIRONMENT

The Server and Client virtual machines were connected through a VirtualBox Host-Only network to allow communication within the lab environment.

The Windows Server machine provided the domain services required by the client, including Active Directory, DNS, and DHCP.

NETWORK ARCHITECTURE

Windows Server (Sv02)



Active Directory / DNS / DHCP

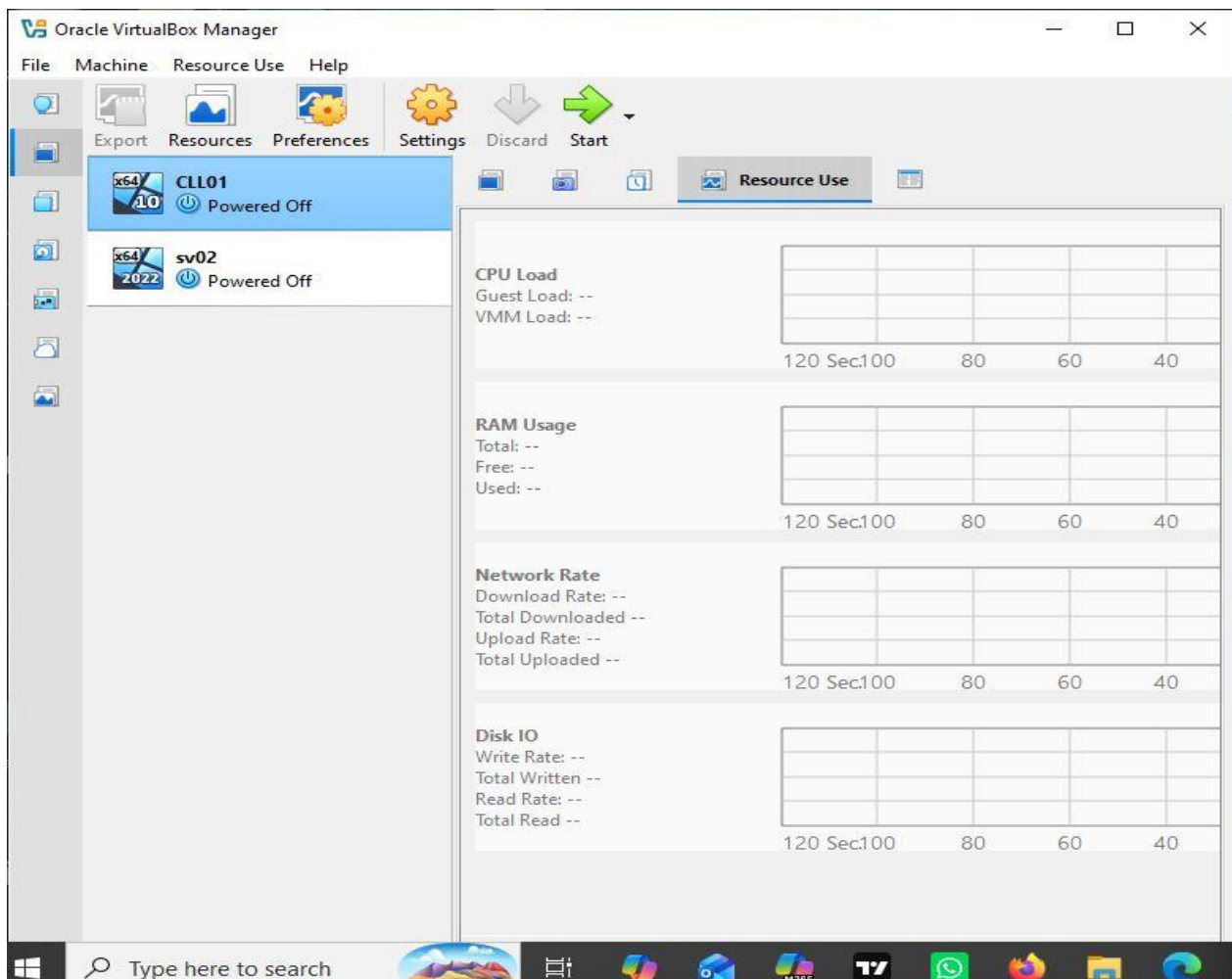


Windows 10 Pro Client (CLL01)



Domain: company.local

This configuration provided an isolated environment for testing Windows Server administration and domain-based management without affecting the host computer's primary network.



2. ACTIVE DIRECTORY CONFIGURATION

DOMAIN CONFIGURATION

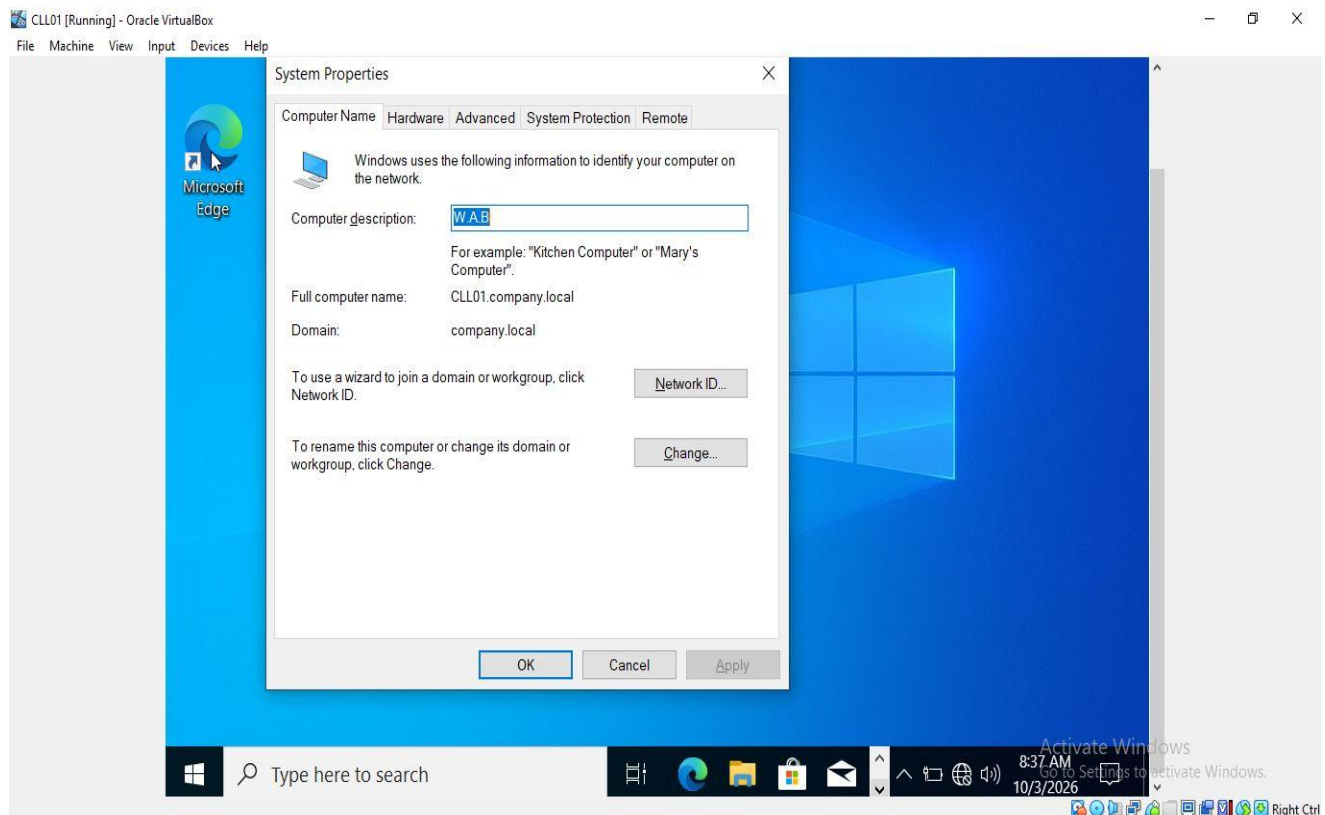
Active Directory Domain Services (AD DS) was configured on the Windows Server virtual machine to establish a centralized domain environment.

The server was configured as a Domain Controller for the domain:

Domain: company.local

Server: Sv02

The domain provides centralized authentication and management of users, a computer, security groups, and Group Policy within the lab environment.



The Windows 10 Pro client was subsequently joined to the company.local domain, confirming communication between the client and the Domain Controller.

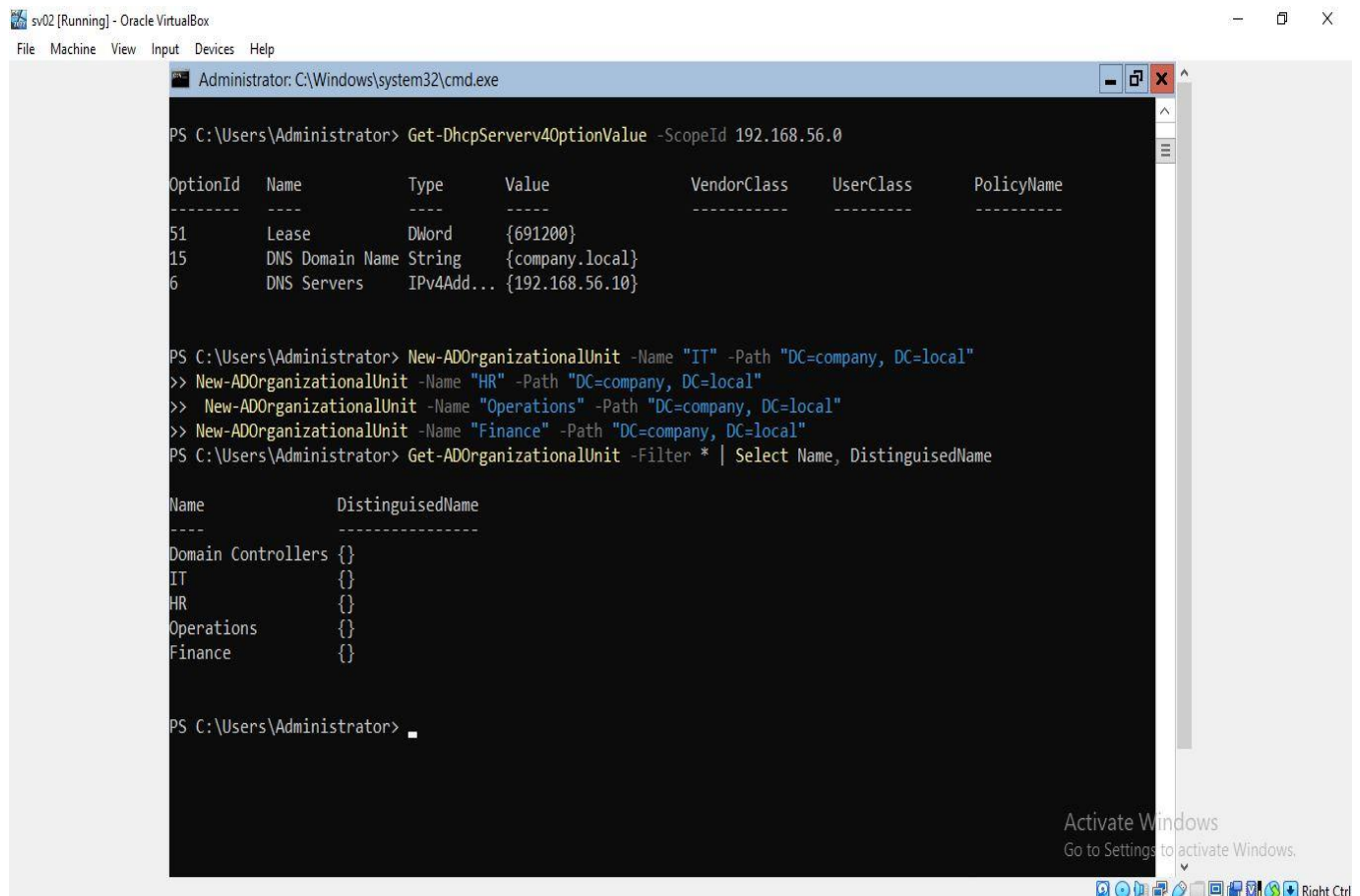
ORGANIZATIONAL UNITS

Organizational Units (OUs) were created in Active Directory to logically organize users according to their respective departments.

The lab included OUs for:

- **Finance**
- **HR**
- **Operations**
- **IT**

This structure provides a foundation for applying department-specific policies, permissions, and administrative controls.



```
Administrator: C:\Windows\system32\cmd.exe

PS C:\Users\Administrator> Get-DhcpServerv4OptionValue -ScopeId 192.168.56.0

OptionId  Name      Type      Value      VendorClass  UserClass  PolicyName
-----
51      Lease     DWord     {691200}
15      DNS Domain Name String    {company.local}
6       DNS Servers IPv4Add... {192.168.56.10}

PS C:\Users\Administrator> New-ADOrganizationalUnit -Name "IT" -Path "DC=company, DC=local"
>> New-ADOrganizationalUnit -Name "HR" -Path "DC=company, DC=local"
>> New-ADOrganizationalUnit -Name "Operations" -Path "DC=company, DC=local"
>> New-ADOrganizationalUnit -Name "Finance" -Path "DC=company, DC=local"
PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Select Name, DistinguishedName

Name      DistinguishedName
-----
Domain Controllers {}
IT         {}
HR         {}
Operations {}
Finance    {}

PS C:\Users\Administrator>
```

The Organizational Units were verified in Active Directory to confirm that the departmental structure was correctly created.

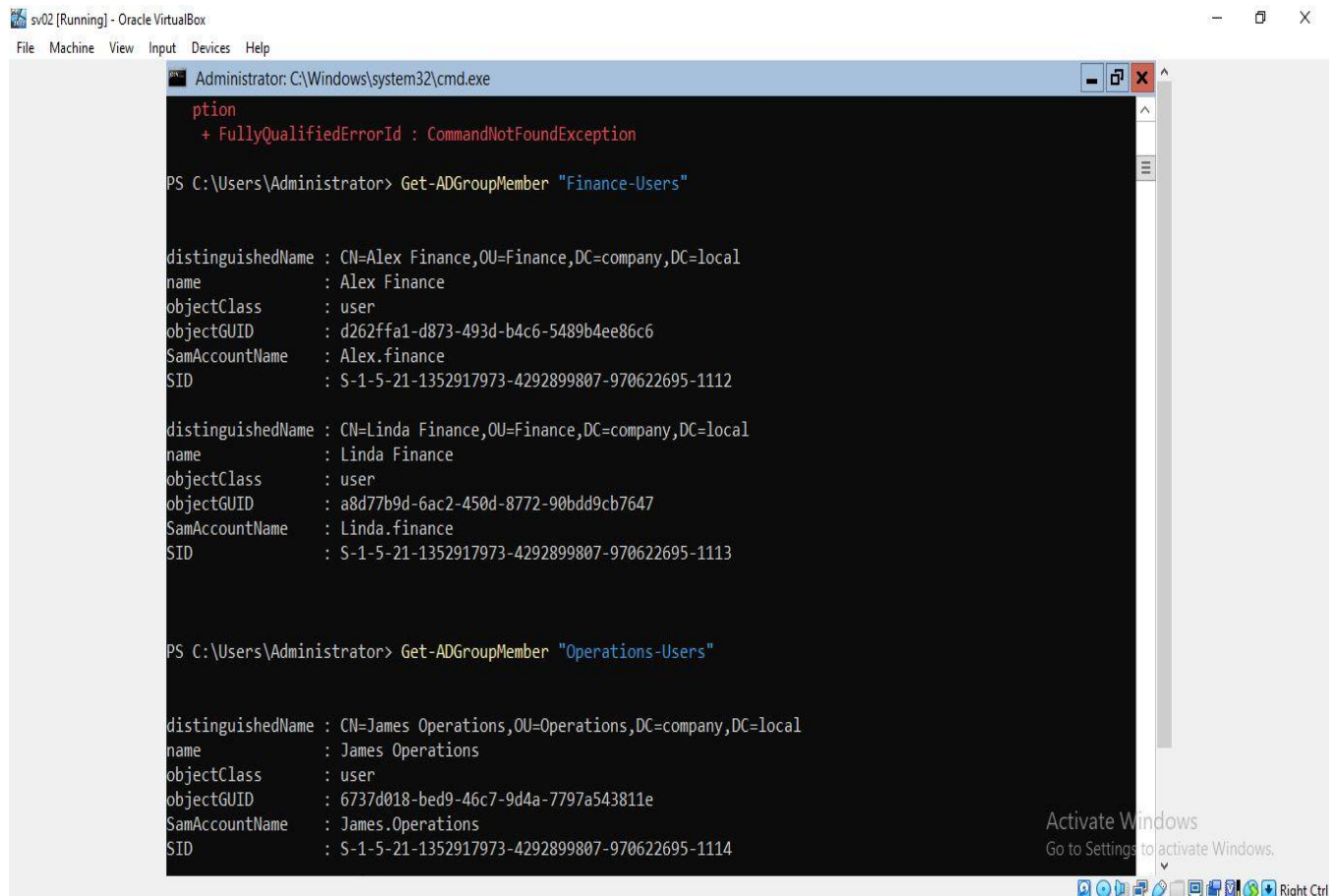
USERS AND SECURITY GROUPS

User accounts and security groups were configured in Active Directory to support centralized identity and access management.

Users were organized according to their departments, while security groups were used to assign access to shared resources based on job function.

Key security groups included:

- **IT-Admins** — administrative access
- **Operations-Users** — Operations department access
- **Finance-Users** — Finance department access.



```
sv02 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Administrator: C:\Windows\system32\cmd.exe

ption
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator> Get-ADGroupMember "Finance-Users"

distinguishedName : CN=Alex Finance,OU=Finance,DC=company,DC=local
name               : Alex Finance
objectClass        : user
objectGUID         : d262ffa1-d873-493d-b4c6-5489b4ee86c6
SamAccountName     : Alex.finance
SID                : S-1-5-21-1352917973-4292899807-970622695-1112

distinguishedName : CN=Linda Finance,OU=Finance,DC=company,DC=local
name               : Linda Finance
objectClass        : user
objectGUID         : a8d77b9d-6ac2-450d-8772-90bdd9cb7647
SamAccountName     : Linda.finance
SID                : S-1-5-21-1352917973-4292899807-970622695-1113

PS C:\Users\Administrator> Get-ADGroupMember "Operations-Users"

distinguishedName : CN=James Operations,OU=Operations,DC=company,DC=local
name               : James Operations
objectClass        : user
objectGUID         : 6737d018-bed9-46c7-9d4a-7797a543811e
SamAccountName     : James.Operations
SID                : S-1-5-21-1352917973-4292899807-970622695-1114
```

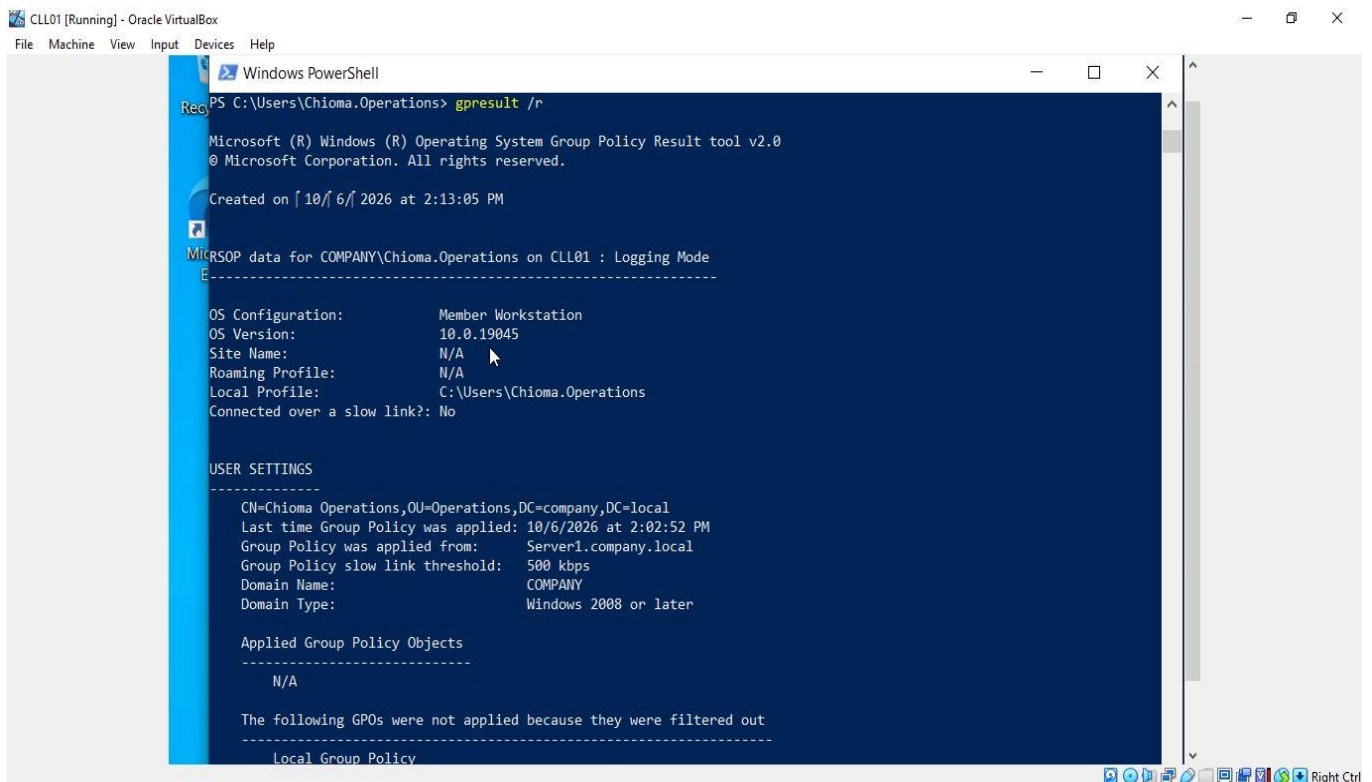
Group membership was verified using Active Directory PowerShell commands to confirm that the appropriate users were assigned to their respective security groups.

DOMAIN JOIN & VERIFICATION

The Windows 10 Pro client, CLL01, was joined to the company.local Active Directory domain.

After joining the domain, domain user accounts were used to authenticate on the client workstation. Group Policy and domain membership were subsequently verified from the client.

This shows successful communication between the Windows 10 client and the Active Directory Domain Controller.



```
PS C:\Users\Chioma.Operations> gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 10/6/2026 at 2:13:05 PM

MIGRATED data for COMPANY\Chioma.Operations on CLL01 : Logging Mode
-----
OS Configuration:      Member Workstation
OS Version:            10.0.19045
Site Name:             N/A
Roaming Profile:       N/A
Local Profile:         C:\Users\Chioma.Operations
Connected over a slow link?: No

USER SETTINGS
-----
CN=Chioma Operations,OU=Operations,DC=company,DC=local
Last time Group Policy was applied: 10/6/2026 at 2:02:52 PM
Group Policy was applied from:      Server1.company.local
Group Policy slow link threshold:   500 kbps
Domain Name:                       COMPANY
Domain Type:                       Windows 2008 or later

Applied Group Policy Objects
-----
N/A

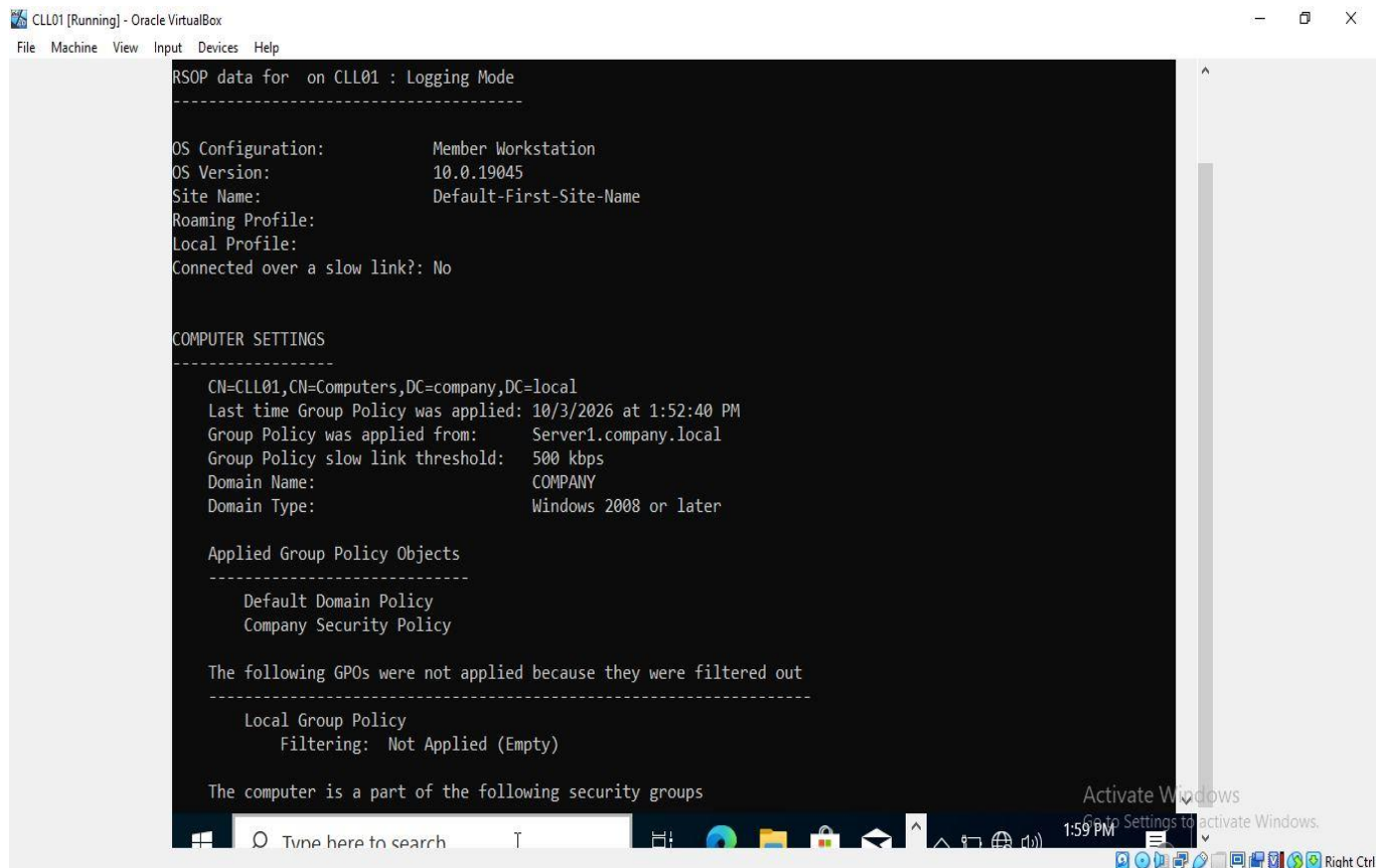
The following GPOs were not applied because they were filtered out
-----
Local Group Policy
```

The client successfully displayed the company.local domain and applied the configured domain Group Policy Objects.

3. GROUP POLICY CONFIGURATION

A Group Policy Object (GPO) named **Company Security Policy** was created and linked to the domain to provide centralized security management for domain-joined workstations.

The policy was applied to the Windows 10 Pro client, CLL01, through the company.local domain.



```
CLL01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

RSOP data for on CLL01 : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.19045
Site Name:             Default-First-Site-Name
Roaming Profile:
Local Profile:
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=CLL01,CN=Computers,DC=company,DC=local
Last time Group Policy was applied: 10/3/2026 at 1:52:40 PM
Group Policy was applied from:    Server1.company.local
Group Policy slow link threshold: 500 kbps
Domain Name:                     COMPANY
Domain Type:                     Windows 2008 or later

Applied Group Policy Objects
-----
Default Domain Policy
Company Security Policy

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The computer is a part of the following security groups

Activate Windows
Go to Settings to activate Windows.
1:59 PM
Type here to search
Right Ctrl
```

Group Policy application was verified on the client using: `gpresult /r`

The results confirmed that **Company Security Policy** and the **Default Domain Policy** were successfully applied to the workstation.

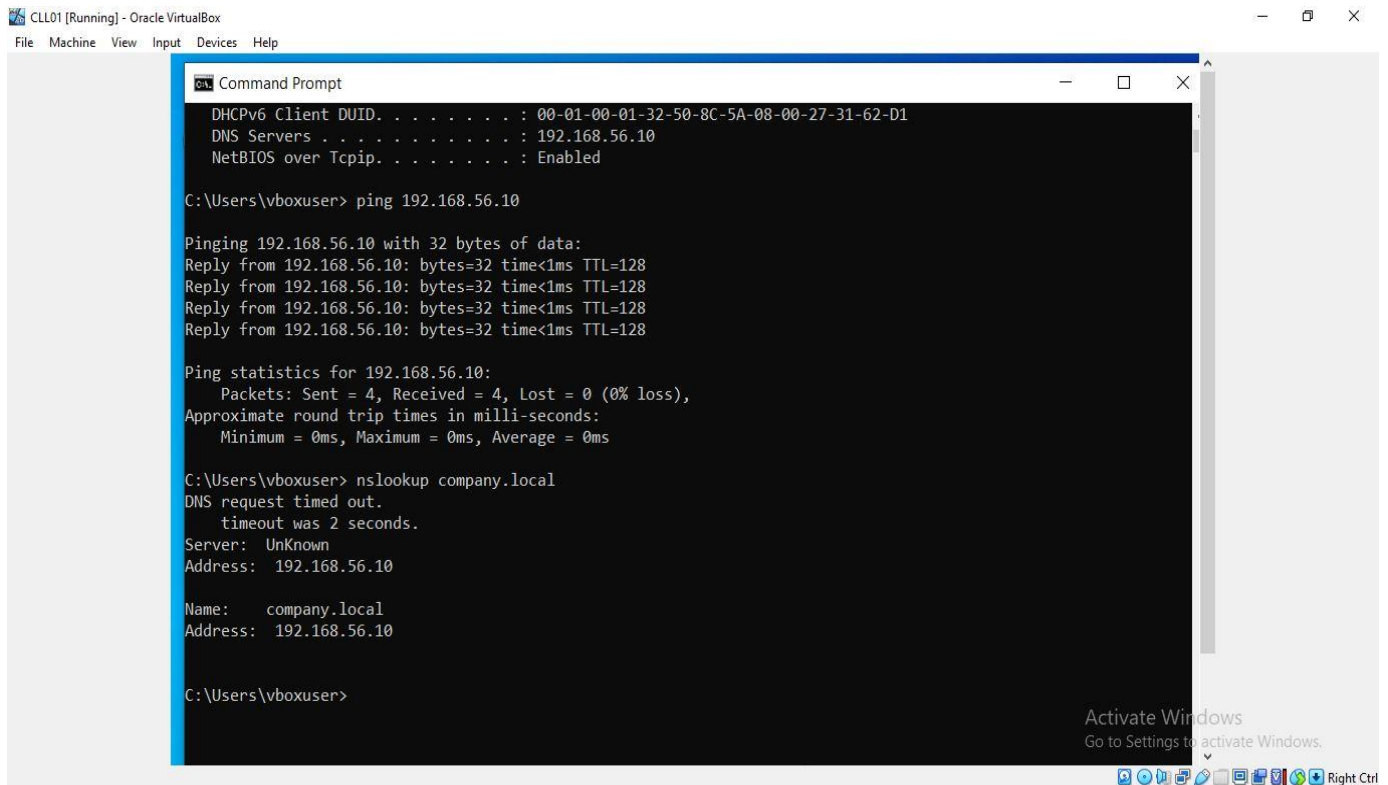
4.

DNS AND DHCP

DNS CONFIGURATION

DNS was configured as part of the Active Directory domain environment to provide name resolution between the Windows Server Domain Controller and domain-joined client.

The Windows 10 client was configured to use the Domain Controller as its DNS server. This allowed the client to resolve the company.local domain and locate domain services.



The screenshot shows a Windows 10 virtual machine window titled "CLL01 [Running] - Oracle VirtualBox". Inside, a Command Prompt window is open, displaying the following output:

```
CLL01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

C:\> ipconfig /all

DHCPv6 Client DUID. . . . . : 00-01-00-01-32-50-8C-5A-08-00-27-31-62-D1
DNS Servers . . . . . : 192.168.56.10
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\vboxuser> ping 192.168.56.10

Pinging 192.168.56.10 with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.56.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\vboxuser> nslookup company.local
DNS request timed out.
    timeout was 2 seconds.
Server:      UnKnown
Address:     192.168.56.10

Name:   company.local
Address: 192.168.56.10

C:\Users\vboxuser>
```

An "Activate Windows" watermark is visible in the bottom right corner of the Command Prompt window.

DNS resolution was tested from the client using: `nslookup company.local`

The domain resolved successfully, confirming that the client could communicate with the configured DNS service.

DHCP CONFIGURATION

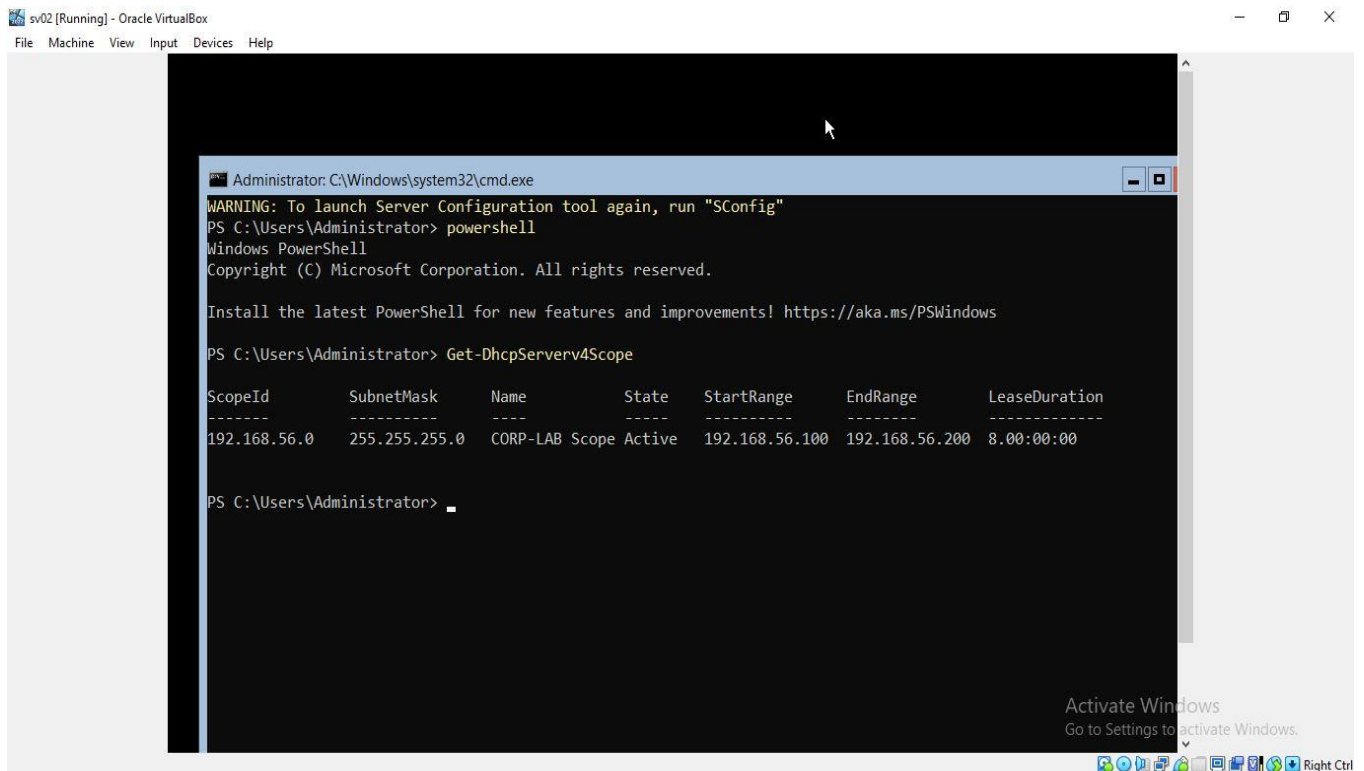
A DHCP scope was configured on the Windows Server to provide automatic IP address assignment to devices within the lab network.

The Windows 10 Pro client was configured to obtain its IP address and DNS settings automatically from the network.

This reduced the need for manual IP configuration on the client and demonstrated centralized network address management.

Verification

The DHCP scope was verified on the Server, and the client successfully obtained an IP address automatically.



The screenshot shows a Windows Server command prompt window titled "Administrator: C:\Windows\system32\cmd.exe". The prompt displays the output of the command `Get-DhcpServerv4Scope`, which returns a table of DHCP scopes. The table has columns for ScopeId, SubnetMask, Name, State, StartRange, EndRange, and LeaseDuration. The output shows a single scope with ID 192.168.56.0, subnet mask 255.255.255.0, name CORP-LAB Scope, and state Active. The start range is 192.168.56.100 and the end range is 192.168.56.200. The lease duration is 8.00:00:00. The prompt also shows a warning message about launching the Server Configuration tool again, and a PowerShell prompt.

```
Administrator: C:\Windows\system32\cmd.exe
WARNING: To launch Server Configuration tool again, run "SConfig"
PS C:\Users\Administrator> powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-DhcpServerv4Scope

ScopeId      SubnetMask    Name          State      StartRange    EndRange      LeaseDuration
-----
192.168.56.0  255.255.255.0 CORP-LAB Scope Active     192.168.56.100 192.168.56.200 8.00:00:00

PS C:\Users\Administrator>
```

5. FILE SERVER AND PERMISSIONS

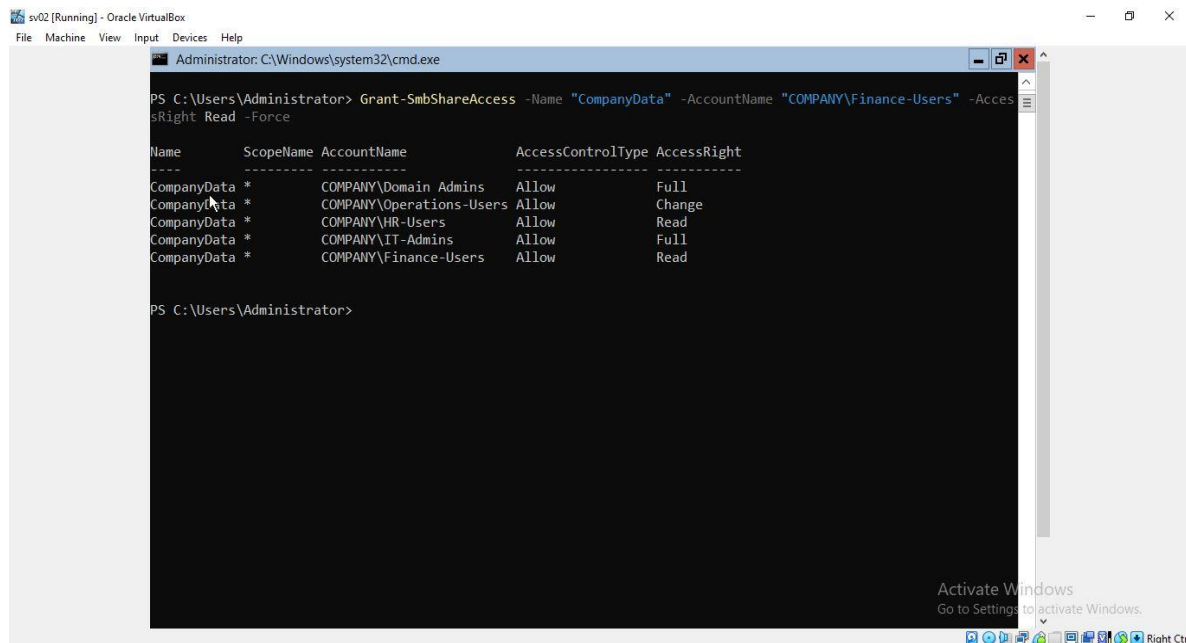
SMB FILE SHARE

An SMB file share named “**CompanyData**” was created on the Windows Server to provide centralized access to shared business data.

Access to the share was controlled using both **Share Permissions** and **NTFS Permissions**, allowing different levels of access to be assigned to specific security groups.

The following access structure was implemented:

Security Group	Access Level
IT-Admins	Full Control
Operations-Users	Change / Modify
HR-Users	Read



```
Administrator: C:\Windows\system32\cmd.exe
PS C:\Users\Administrator> Grant-SmbShareAccess -Name "CompanyData" -AccountName "COMPANY\Finance-Users" -AccessRight Read -Force

Name      ScopeName AccountName      AccessControlType AccessRight
-----
CompanyData *      COMPANY\Domain Admins      Allow      Full
CompanyData *      COMPANY\Operations-Users    Allow      Change
CompanyData *      COMPANY\HR-Users           Allow      Read
CompanyData *      COMPANY\IT-Admins          Allow      Full
CompanyData *      COMPANY\Finance-Users       Allow      Read

PS C:\Users\Administrator>
```

This configuration demonstrates role-based access control for shared network resources.

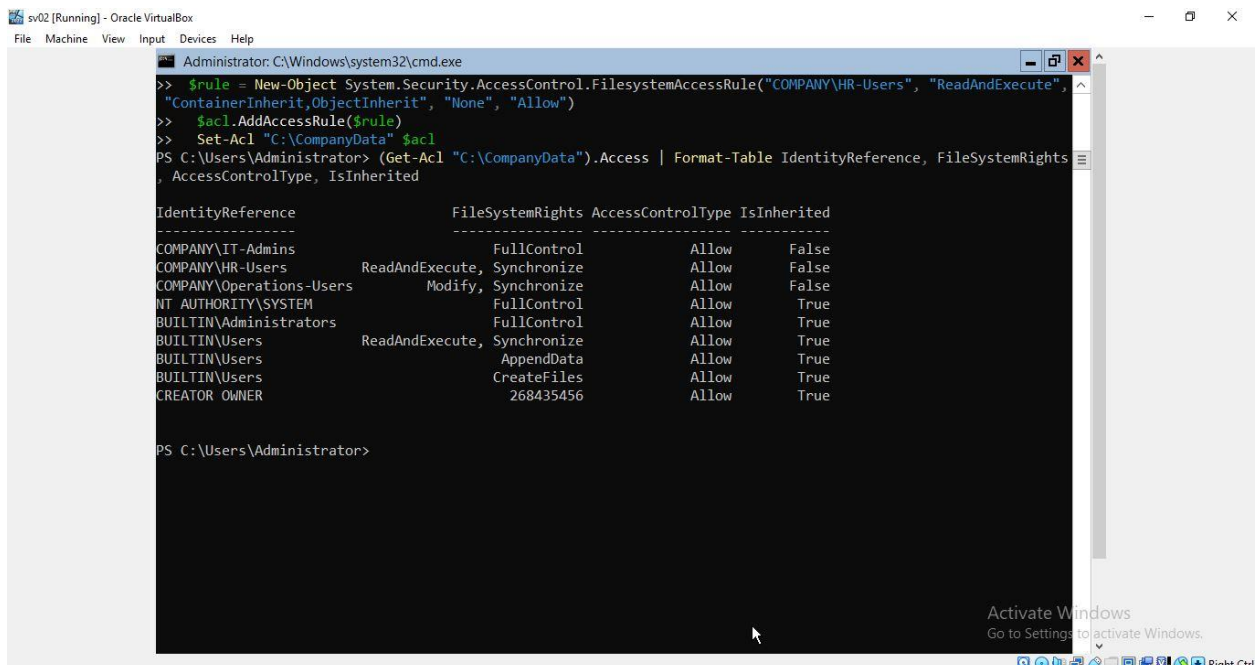
NTFS Permissions

NTFS permissions were configured on the **CompanyData** folder to provide granular access control based on departmental security groups.

The configured permissions were:

Security Group	NTFS Permission
IT-Admins	Full Control
Operations-Users	Modify
HR-Users	Read & Execute

These permissions were configured separately from the SMB share permissions to provide an additional layer of access control.



```
Administrator: C:\Windows\system32\cmd.exe
>> $rule = New-Object System.Security.AccessControl.FileSystemAccessRule("COMPANY\HR-Users", "ReadAndExecute",
"ContainerInherit, ObjectInherit", "None", "Allow")
>> $acl.AddAccessRule($rule)
>> Set-Acl "C:\CompanyData" $acl
PS C:\Users\Administrator> (Get-Acl "C:\CompanyData").Access | Format-Table IdentityReference, FileSystemRights
, AccessControlType, IsInherited

IdentityReference      FileSystemRights AccessControlType IsInherited
-----
COMPANY\IT-Admins      FullControl       Allow             False
COMPANY\HR-Users       ReadAndExecute, Synchronize Allow             False
COMPANY\Operations-Users Modify, Synchronize Allow             False
NT AUTHORITY\SYSTEM    FullControl       Allow             True
BUILTIN\Administrators FullControl       Allow             True
BUILTIN\Users          ReadAndExecute, Synchronize Allow             True
BUILTIN\Users          AppendData        Allow             True
BUILTIN\Users          CreateFiles       Allow             True
CREATOR OWNER          268435456        Allow             True

PS C:\Users\Administrator>
```

The NTFS access control entries were verified using PowerShell to confirm that the appropriate permissions were assigned to each security group.

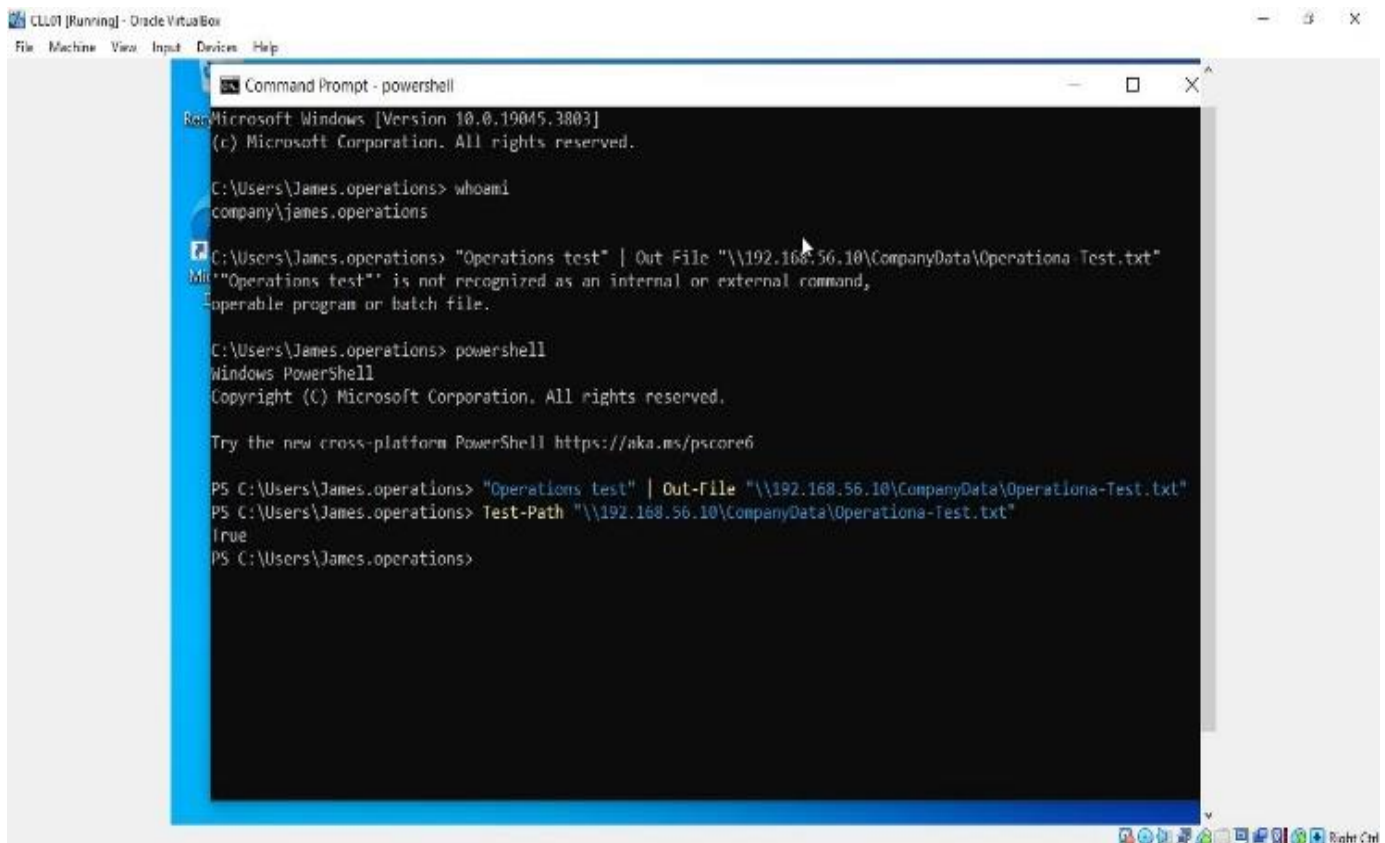
ACCESS TESTING

The configured file-share permissions were tested using domain accounts belonging to the IT, Operations, and HR security groups.

The tests confirmed that access was granted according to each group's assigned permissions:

- **IT-Admins:** Successfully created a test file, confirming write access.
- **Operations-Users:** Successfully created a test file, confirming write access.
- **HR-Users:** Successfully accessed existing files but was prevented from creating new files, confirming read-only access.

These tests demonstrated that the implemented Share and NTFS permissions were functioning as intended.



```
CLLO1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Command Prompt - powershell
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\James.operations> whoami
company\james.operations

C:\Users\James.operations> "Operations test" | Out-File "\\192.168.56.10\CompanyData\Operationa-Test.txt"
Microsoft "Operations test" is not recognized as an internal or external command,
operable program or batch file.

C:\Users\James.operations> powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\James.operations> "Operations test" | Out-File "\\192.168.56.10\CompanyData\Operationa-Test.txt"
PS C:\Users\James.operations> Test-Path "\\192.168.56.10\CompanyData\Operationa-Test.txt"
True
PS C:\Users\James.operations>
```

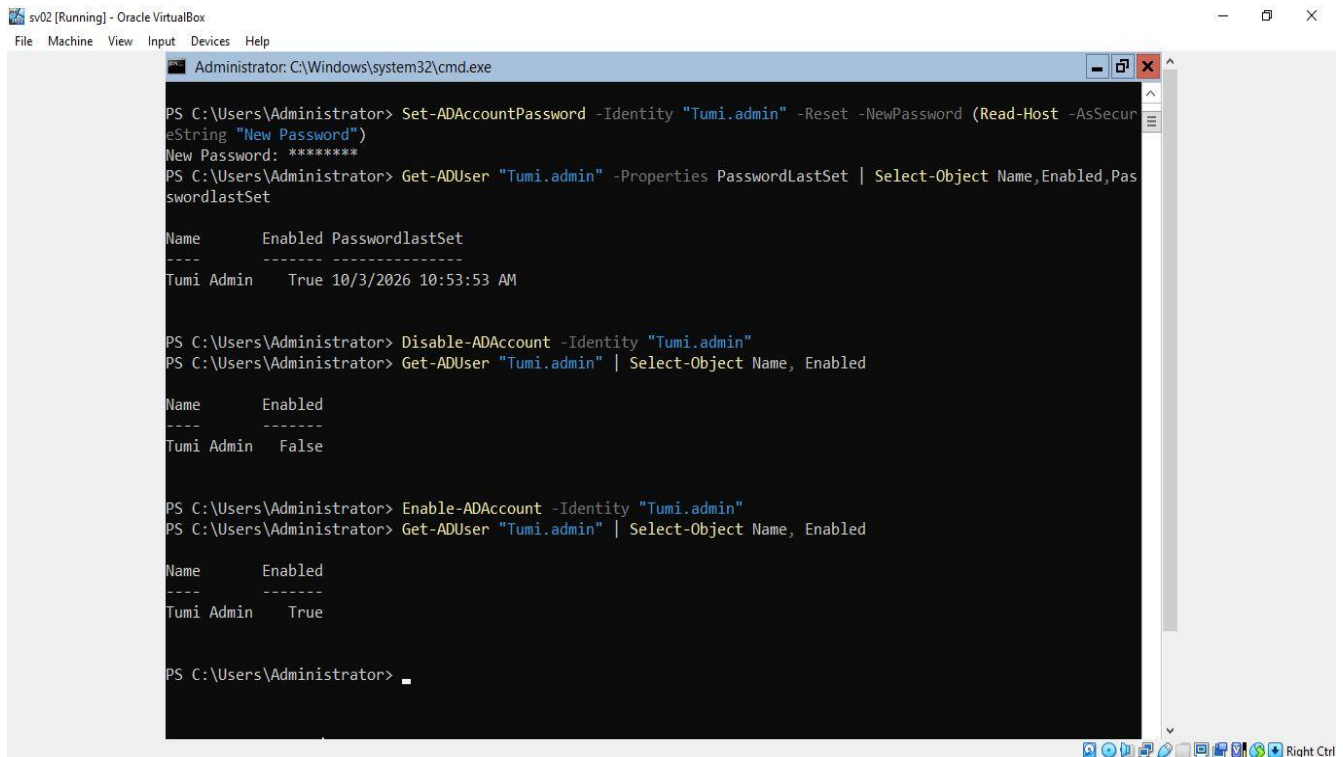

6. ACCOUNT ADMINISTRATION

Basic user account lifecycle administration was performed using Active Directory PowerShell.

An existing domain user account was temporarily disabled and verified to confirm that the account status changed to disabled. The account was subsequently re-enabled and verified again to confirm that it was active.

The following PowerShell cmdlets were used:

- Disable-ADAccount
- Get-ADUser
- Enable-ADAccount



```
Administrator: C:\Windows\system32\cmd.exe

PS C:\Users\Administrator> Set-ADAccountPassword -Identity "Tumi.admin" -Reset -NewPassword (Read-Host -AsSecureString "New Password")
New Password: *****

PS C:\Users\Administrator> Get-ADUser "Tumi.admin" -Properties PasswordLastSet | Select-Object Name,Enabled,PasswordLastSet

Name           Enabled PasswordLastSet
-----
Tumi Admin     True  10/3/2026 10:53:53 AM

PS C:\Users\Administrator> Disable-ADAccount -Identity "Tumi.admin"
PS C:\Users\Administrator> Get-ADUser "Tumi.admin" | Select-Object Name, Enabled

Name           Enabled
-----
Tumi Admin     False

PS C:\Users\Administrator> Enable-ADAccount -Identity "Tumi.admin"
PS C:\Users\Administrator> Get-ADUser "Tumi.admin" | Select-Object Name, Enabled

Name           Enabled
-----
Tumi Admin     True

PS C:\Users\Administrator> _
```

This demonstrated the ability to manage the status of domain user accounts through Active Directory.

7. TESTING & RESULTS

The completed Windows Server environment was tested to confirm that the configured services, policies, and access controls were functioning as intended.

The results confirmed that the Windows Server environment was operational and that users received access according to their assigned roles and permissions.

Test	Result
Windows 10 client joined to company.local	Successful
Domain authentication	Successful
DNS resolution	Successful
DHCP address assignment	Successful
Group Policy application	Successful
SMB share accessibility	Successful
IT-Admins write access	Successful
Operations-Users write access	Successful
HR read-only access	Successful
User account disable/re-enable	Successful

8. SKILLS DEMONSTRATED

This project provided hands-on experience with the following technologies and IT administration tasks:

- **Windows Server Administration**
- **Active Directory Domain Services (AD DS)**
- **User, Group and Organizational Unit Management**
- **Group Policy Management**
- **DNS Configuration and Troubleshooting**
- **DHCP Configuration**
- **Windows 10 Domain Joining**
- **SMB File Sharing**
- **NTFS and Share Permissions**
- **Role-Based Access Control**
- **Active Directory User Account Administration**
- **PowerShell for Windows Administration**
- **Network and Access Troubleshooting**
- **System and Configuration Verification**

The project demonstrates practical experience in configuring, managing, testing, and troubleshooting a small Windows-based business IT environment.